Strategy Design Pattern

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Defn

1. Behavioral Design Pattern
2. The strategy design pattern is to define a set of algorithms, encapsulate them in their classes, and make them interchangable within content objects.

Where to use ?

=> If we have functionality which can be implemented in different ways, then we can use strategy design pattern.

Example

```java
class GoogleMapClient {
    Run | Debug
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        String mode = sc.next();

        if(mode.equals(anObject: "car")) {

            // Logic to find Car path
            // 100 lines of code

        } else if(mode.equals(anObject: "walk")) {

            // Logic to find Car path
            // 100 lines of code

        } else {
            throw new IllegalArgumentException(s: "Invalid Input");
        }
    }
}
```

Problems

→ Too many if-else

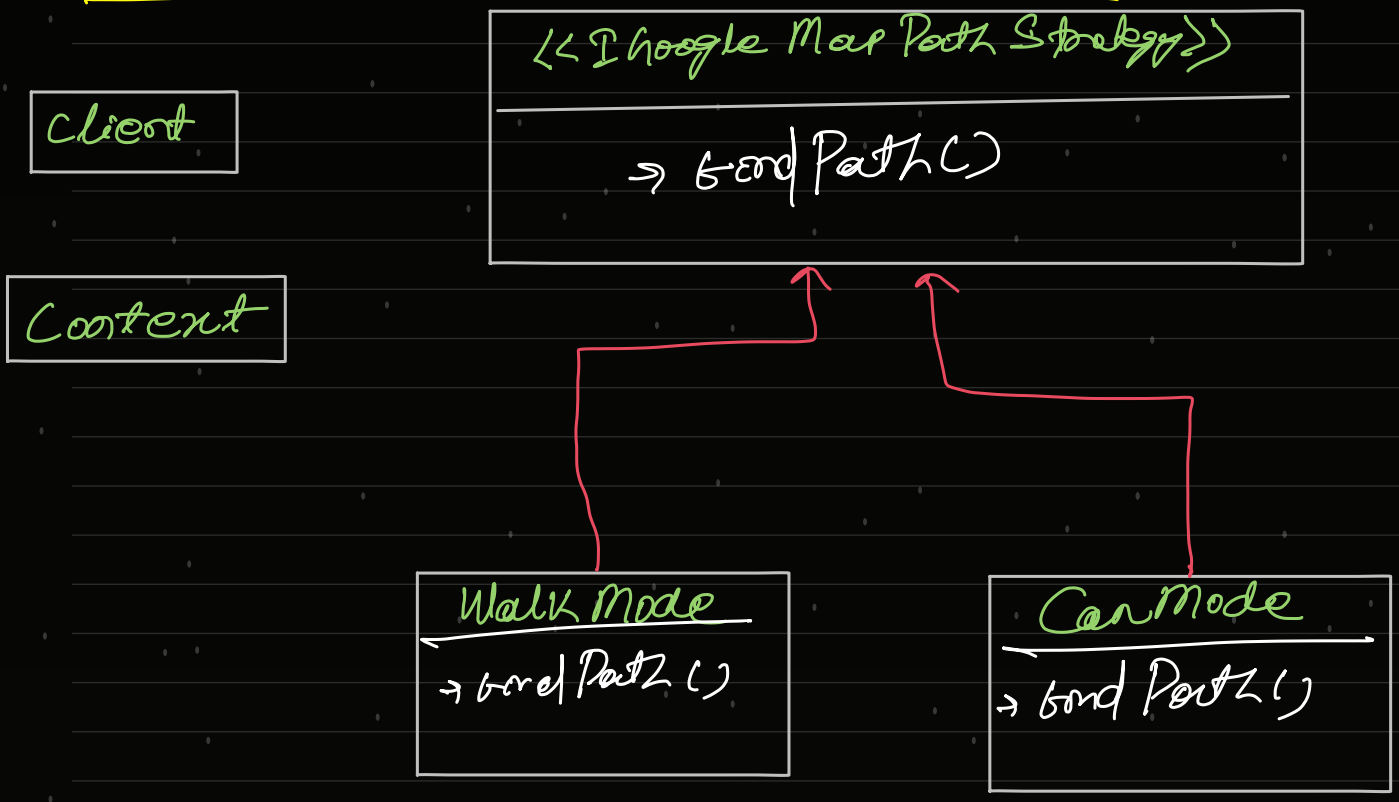
→ Hard to extend

→ Violates

→ SRP

→ OCP

How to use strategy design pattern



Strategy	VS	Adapter
① Behavioral		① Structural
② If we have functionality which can be implemented in different ways, then we can use strategy design pattern.		② If there is a requirement where you need to integrate 3rd party SDK or library then we can use adapter design pattern.